

Prime Pairs at the Prime Number Theorem Threshold

Hardy–Littlewood for Almost Every Shift in Every Interval of Length $X^{2/15+\varepsilon}$

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Abstract

Let

$$S_X(\alpha) = \sum_{X < n \leq 2X} \Lambda(n) e^{2\pi i n \alpha}.$$

For every $A, \varepsilon > 0$, we choose narrow neighborhoods of rational points with logarithmically bounded denominators and prove

$$\sup_{|J|=1/H} \int_{J \setminus \mathfrak{M}} |S_X(\alpha)|^2 d\alpha \ll_{A,\varepsilon} X (\log X)^{-A}$$

whenever $H \geq X^{2/15+\varepsilon}$. It follows that, in every translated shift interval of the stated length, the Hardy–Littlewood prime-pair formula holds in mean square and for all but $O(H(\log X)^{-A})$ individual shifts, and the second moment of R_X matches its singular-series prediction. The same estimate gives the weighted Goldbach formula for almost every target in $[X, X+H]$, together with a balanced version whose summands lie in $(X, 2X]$. This replaces the exponent $8/33$ in Matomäki–Radziwiłł–Tao by $2/15$.

Introduction

The prime number theorem in short intervals counts one copy of the primes:

$$\psi(x+y) - \psi(x) \sim y.$$

The Hardy–Littlewood problem correlates the primes with a translate:

$$R_X(h) = \sum_{X < n \leq 2X} \Lambda(n) \Lambda(n+h) \sim X \mathfrak{S}(h).$$

The singular series $\mathfrak{S}(h)$ is the product of the local congruence factors. At $h = 2$, this correlation contains the twin-prime problem. The theorems below are averages over h and leave every fixed shift open. Averaging in h lets us reach the $2/15$ short-interval scale.

The current one-point theory gives the prime number theorem for almost all intervals of length $x^{2/15+\varepsilon}$ [2]. Matomäki, Radziwiłł, and Tao [8] proved the Hardy–Littlewood prime-pair asymptotic for almost all shifts in families of length $X^{8/33+\varepsilon}$. Theorem 1.1 lowers the two-point threshold to $X^{2/15+\varepsilon}$, so the one-point and two-point thresholds now agree. For $X = 10^{30}$, the old and new shift scales are approximately $10^{7.27}$ and 10^4 : about nineteen million shifts versus ten thousand.

The estimate is uniform in the center of the shift interval: in every translated interval of length $X^{2/15+\varepsilon}$, almost all shifts obey the Hardy–Littlewood prediction.

1 Statements

Write $L = \log X$, $e(t) = e^{2\pi it}$, and identify $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For positive P, Q set

$$\mathfrak{M}(P, Q) = \bigcup_{q \leq Q} \bigcup_{\substack{a \bmod q \\ (a, q) = 1}} \left\{ \alpha \in \mathbb{T} : \left\| \alpha - \frac{a}{q} \right\|_{\mathbb{T}} \leq \frac{P}{X} \right\}, \quad \mathfrak{m}(P, Q) = \mathbb{T} \setminus \mathfrak{M}(P, Q).$$

Theorem 1.1 (Uniform local minor-arc estimate). *For every $A > 0$ and $\varepsilon > 0$ there are positive integers $B = B(A, \varepsilon)$, $D = D(A, \varepsilon)$ and positive constants $C = C(A, \varepsilon)$ and $X_0 = X_0(A, \varepsilon)$ such that, for $X \geq X_0$, with*

$$Q = L^B, \quad P = L^D,$$

one has, simultaneously for every real $H \geq X^{2/15+\varepsilon}$,

$$\sup_{\substack{J \subset \mathbb{T} \text{ an arc} \\ |J| = 1/H}} \int_{J \setminus \mathfrak{M}(P, Q)} |S_X(\alpha)|^2 d\alpha \leq CXL^{-A}. \quad (1.1)$$

The order of choice is

$$A, \varepsilon \longrightarrow B, D, C, X_0 \longrightarrow X, H, J.$$

The constants may be ineffective.

Remark 1.2 (Formalization). The estimate (1.1) is proved in Lean 4 as `AllCenterMAP.map_two_fifteenths`. See <https://github.com/jconorgrogan/prime-minor-arcs-2-15>; the formalization has been submitted to Palomar.

For $h \neq 0$ define

$$R_X(h) = \sum_{X < n \leq 2X} \Lambda(n)\Lambda(n+h)$$

and

$$\mathfrak{S}(h) = \begin{cases} 2C_2 \prod_{\substack{p|h \\ p>2}} \frac{p-1}{p-2}, & 2 \mid h, \\ 0, & 2 \nmid h, \end{cases} \quad C_2 = \prod_{p>2} \frac{p(p-2)}{(p-1)^2}.$$

Theorem 1.3 (Translated prime-pair consequences). *Fix $A, \varepsilon > 0$. Uniformly for*

$$X^{2/15+\varepsilon} \leq H \leq X^{1-\varepsilon}, \quad 0 \leq h_0 \leq X^{1-\varepsilon},$$

and $I = \{h \in \mathbb{Z} : |h - h_0| \leq H, h \neq 0\}$,

$$\sum_{h \in I} |R_X(h) - X\mathfrak{S}(h)|^2 \ll_{A, \varepsilon} HX^2L^{-A}, \quad (1.2)$$

$$\sum_{h \in I} R_X(h)^2 = X^2 \sum_{h \in I} \mathfrak{S}(h)^2 + O_{A, \varepsilon}(HX^2L^{-A}), \quad (1.3)$$

and

$$\#\{h \in I : |R_X(h) - X\mathfrak{S}(h)| > XL^{-A}\} \ll_{A, \varepsilon} HL^{-A}. \quad (1.4)$$

For an integer $N \geq 2$, let

$$G(N) = \sum_{n+m=N} \Lambda(n)\Lambda(m),$$

where the pairs are ordered, and let $\mathfrak{S}(N)$ denote the binary Goldbach singular series, equal to zero for odd N and to the expression above for even N .

Theorem 1.4 (Averaged Goldbach at 2/15). *For every $A, \varepsilon > 0$, uniformly for*

$$X^{2/15+\varepsilon} \leq H \leq X^{1-\varepsilon},$$

one has

$$G(N) = \mathfrak{S}(N)N + O_{A,\varepsilon}(XL^{-A}) \quad (1.5)$$

for all but $O_{A,\varepsilon}(HL^{-A})$ integers $N \in [X, X+H]$. The same formula holds for the log-weighted prime-only sum

$$W(N) = \sum_{\substack{p+q=N \\ p,q \text{ prime}}} \log p \log q.$$

Every sufficiently large even target outside the exceptional set has a Goldbach representation.

2 Two estimates

The first estimate integrates the two Dirichlet polynomials in t and in their relative shift u at the same time. The second gives the prime number theorem simultaneously for every short interval and residue class, for all moduli $q \leq (\log X)^K$.

2.1 A two-variable mean for Dirichlet polynomials

Lemma 2.1 (Mixed mean). *Let $c > 0$, let $a \geq 0$ be an integer, and let $k \geq 1$ be an integer. Let $M, N \geq 2$, $N \geq cM^2$, and $T, U \geq 1$. Suppose $|\beta(m)| \leq \log^a(2m)$ on $(M, 2M]$ and $|g(n)| \leq \tau_k(n) \log^a(2n)$ on $(N, 2N]$. Put*

$$B(t) = \sum_m \beta(m) m^{-1/2-it}, \quad G(t) = \sum_n g(n) n^{-1/2-it}.$$

If I is any interval of length T , then, with $R = \max(2, k^2)$,

$$\int_I |G(t)|^2 \int_{|u| \leq 2U} |B(t+u)|^2 du dt \ll_{c,a,k} \log^{4a+2R+2}(2MN)(UT + UN + MN + T). \quad (2.1)$$

Fixed character phases may be inserted in either polynomial.

Proof. Use a fixed multiple of

$$\eta((t-t_0)/T)\eta(u/U), \quad \eta(v) = \left(\frac{\sin(\pi v/4)}{\pi v/4} \right)^2,$$

to majorize the rectangle. The Fourier transform of η is supported in $[-1/4, 1/4]$. Expanding the two squares therefore leaves only quadruples with

$$|m_1 - m_2| \leq K := \min(M, \pi M/U), \quad |m_1 n_1 - m_2 n_2| \leq Y := 2\pi MN/T.$$

The four weights $m_i^{-1/2}n_i^{-1/2}$ contribute $O((MN)^{-1})$, and the two Fourier transforms contribute $O(TU)$. The left side of (2.1) is at most a fixed power of $\log(2MN)$ times

$$\frac{TU}{MN} \sum_{\substack{m_i \asymp M, n_i \asymp N \\ |m_1 - m_2| \leq K, |m_1 n_1 - m_2 n_2| \leq Y}} \tau_k(n_1) \tau_k(n_2). \quad (2.2)$$

We split this sum into three cases.

If $m_1 = m_2$, then $n_1 - n_2 = O(N/T)$. Cauchy's inequality and $\sum_{n \leq 2N} \tau_k(n)^2 \ll N \log^{R-1}(2N)$ give the terms $UT + UN$.

Now let $m_1 \neq m_2$. Write $m_1 = da$, $m_2 = db$, $(a, b) = 1$. For a nonzero determinant,

$$an_1 - bn_2 = \ell, \quad 0 < |\ell| \leq Y/d.$$

Set $d_1 = (\ell, b)$ and $d_2 = (\ell, a)$. Dividing the first congruence by d_1 and the second by d_2 gives reduced residue classes, with moduli $O_c((N/d_i)^{1/2})$. Apply Shiu's theorem [9, Theorem 1] on the full intervals $N < n_i \leq 2N$, followed by Cauchy's inequality. For each fixed ℓ , the result is

$$\frac{Nd}{M} \log^R(2MN) \sqrt{\tau_R(d_1) \tau_R(d_2)}.$$

Since $(a, b) = 1$, $d_1 d_2 = (\ell, ab)$, while

$$\sum_{0 < |\ell| \leq Y/d} \tau_R((\ell, ab)) = 2 \sum_{v|ab} \tau_{R-1}(v) \left\lfloor \frac{Y}{dv} \right\rfloor \ll \frac{Y}{d} \log^{R-1}(2MN). \quad (2.3)$$

If $Y/d < 1$, there is no nonzero value of ℓ . Summing the possible pairs (m_1, m_2) gives the term MN in (2.1).

For determinant zero, $n_1 = br$ and $n_2 = ar$. Divisor submultiplicativity and the divisor second moment give

$$\sum_r \tau_k(br) \tau_k(ar) \ll \frac{Nd}{M} \log^{R-1}(2MN) \tau_k(a) \tau_k(b).$$

Here $d \leq K$. For fixed d , Cauchy's inequality bounds the neighboring a, b pairs by

$$(M/d)(K/d) \log^{R-1}(2MN).$$

Summing $d \leq K$ gives the term T . Combining the three cases proves (2.1). Character twists have modulus at most one throughout. $\square$

2.2 A uniform short-interval prime number theorem

Proposition 2.2 (Simultaneous short-interval progressions). *Fix $D, K > 0$ and $\varepsilon > 0$. For all sufficiently large X ,*

$$\int_{X/2}^{4X} \max_{q \leq L^K} \max_{(a,q)=1} \sup_{X^{2/15+\varepsilon} \leq Y \leq X} \left| \frac{\psi(x+Y; q, a) - \psi(x; q, a) - Y/\varphi(q)}{Y} \right|^2 dx \ll_{D,K,\varepsilon} X L^{-D}. \quad (2.4)$$

The threshold and the implied constant may be ineffective.

Proof. We give the exponent calculation, since it is what fixes $2/15$. It is enough to treat $0 < \varepsilon \leq 1/10$, since a larger value follows by decreasing ε . Put

$$\theta = \frac{2}{15} + \varepsilon, \quad \tau = \frac{13}{15} - \frac{\varepsilon}{2}, \quad T = X^\tau, \quad Q = L^K.$$

For each character χ , use the primitive character that induces it. If $\rho = \beta + i\gamma$ runs over the nontrivial zeros of the corresponding L -function, counted with multiplicity, set

$$Z = \sum_{q \leq Q} \sum_{\chi \bmod q} \sum_{|\gamma| \leq T} X^{2(\beta-1)}.$$

Set $\delta_0 = \min(\varepsilon/16, 1/120)$. The density theorem for polylogarithmic conductors proved in Appendix A, used with exponent $K+1$ since $Q \leq (\log T)^{K+1}$ for large X , gives, for every fixed $\eta_{\text{zd}} > 0$, uniformly on

$$\begin{aligned} \frac{1}{2} + \delta_0 \leq \sigma \leq \frac{4}{5}, \\ \sum_{q \leq Q} \sum_{\chi \bmod q} N(\sigma, T, \chi) \ll_{K, \delta_0, \eta_{\text{zd}}} T^{\frac{30}{13}(1-\sigma) + \eta_{\text{zd}}}. \end{aligned} \quad (2.5)$$

Its inputs are Montgomery's classical Dirichlet- L zero detector and moment estimates together with Guth–Maynard's published arbitrary-coefficient large-values theorem, applied separately to each of the only polylogarithmically many characters. Thus no hybrid character-family theorem is used. The zeros with $\beta < 1/2 + \delta_0$ contribute

$$\ll Q^2 T L X^{-1+2\delta_0} \ll_{\varepsilon, K} X^{-c_\varepsilon}$$

by the elementary zero count. Partition $[1/2 + \delta_0, 4/5]$ into $O_\varepsilon(1)$ intervals $[\sigma, \sigma + \Delta]$ with

$$\Delta \leq \min\left(\frac{1}{100}, \frac{3\varepsilon}{52}\right),$$

and apply (2.5) at each left endpoint. Choose

$$\eta_{\text{zd}} = \frac{3\varepsilon}{52\tau}.$$

Since

$$2 - \frac{30}{13}\tau = \frac{15}{13}\varepsilon > 0,$$

the X -exponent of a mesh cell is at most

$$-\frac{15\varepsilon}{13}(1-\sigma) + 2\Delta + \tau\eta_{\text{zd}} \leq -\frac{3\varepsilon}{52}.$$

Here we used $\sigma \leq 4/5$ and $\Delta \leq 3\varepsilon/52$. Thus every mesh interval contributes a fixed negative power of X , uniformly over all $q \leq Q$.

Near one, Jutila [5, (1.7)] proves, for every fixed $\eta > 0$,

$$\sum_{\chi \bmod q} N(\sigma, T, \chi) \ll_\eta (qT)^{(2+\eta)(1-\sigma)}, \quad 4/5 \leq \sigma < 1. \quad (2.6)$$

With $\eta = 1/10$,

$$2 - \frac{21 \log(qT)}{10 \log X} \geq \frac{1}{6}$$

for large X . Khale's Vinogradov–Korobov region [6, Theorem 1.1] gives, for every zero with $|\gamma| \geq 10$,

$$1 - \beta \gg_K L^{-2/3} (\log L)^{-1/3}.$$

For the conductor-one zeta factor, apply Khale's theorem to the principal character modulo 3; the additional factor $1 - 3^{-s}$ has zeros only on $\Re s = 0$. The classical low-height region and Siegel's theorem handle the possible real exceptional zero; for $q \leq L^K$ its weighted contribution is $O(L^{2K} \exp(-c_K L^{3/4}))$. For every fixed D ,

$$Z \ll_{D,K,\varepsilon} L^{-D}. \quad (2.7)$$

The explicit formula and local zero count used here are [7, Theorem 11.3, Lemma 11.4(a), Theorems 12.3 and 12.10]. They apply to arbitrary characters; changing to the inducing character costs $O(\log q \log X)$.

Let

$$F_\chi(t) = \sum_{|\gamma| \leq T} t^{\rho-1} \quad (X/4 \leq t \leq 6X),$$

extended by zero, where the zeros are those of the primitive character inducing χ . If $\rho = \beta + i\gamma$ and $\rho' = \beta' + i\gamma'$, direct integration gives

$$\left| \int_{X/4}^{6X} t^{\rho+\bar{\rho}'-2} dt \right| \ll \frac{X^{\beta+\beta'-1}}{1+|\gamma-\gamma'|}.$$

Writing $w_\rho = X^{\beta-1}$, using $2w_\rho w_{\rho'} \leq w_\rho^2 + w_{\rho'}^2$, and grouping the ordinates of ρ' into unit intervals, Lemma 11.4(a) and the harmonic sum give

$$\|F_\chi\|_2^2 \ll XL^2 \sum_{|\gamma| \leq T} X^{2(\beta-1)}.$$

Consequently

$$\sum_{q \leq Q} \sum_{\chi \bmod q} \|F_\chi\|_2^2 \ll XL^2 Z. \quad (2.8)$$

Let

$$E_\chi(x, Y) = \psi(x+Y, \chi) - \psi(x, \chi) - \mathbf{1}_{\chi=\chi_0} Y.$$

Subtracting the explicit formula at $x+Y$ and x , and using the primitive–imprimitive correction, gives uniformly in the asserted ranges

$$\frac{|E_\chi(x, Y)|}{Y} \leq M^+ F_\chi(x) + O(X^{-\varepsilon/2} L^2), \quad (2.9)$$

where M^+ is the one-sided Hardy–Littlewood maximal operator. Character orthogonality is used only before the supremum:

$$E_{q,a}(x, Y) = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \bar{\chi}(a) E_\chi(x, Y).$$

After (2.9), Cauchy–Schwarz, rather than orthogonality, yields

$$\sup_Y \left| \frac{E_{q,a}(x, Y)}{Y} \right|^2 \leq \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \left(M^+ F_\chi(x) + O(X^{-\varepsilon/2} L^2) \right)^2. \quad (2.10)$$

The right side is independent of a . Since the summands are nonnegative, $\max_{q \leq Q}$ is at most their sum over q . The L^2 boundedness of M^+ , followed by (2.7)–(2.8), therefore gives

$$\int_{X/2}^{4X} \max_{q \leq Q} \max_{(a,q)=1} \sup_Y \left| \frac{E_{q,a}(x, Y)}{Y} \right|^2 dx \ll XL^2Z + QX^{1-\varepsilon}L^4.$$

Taking the logarithmic saving in (2.7) larger by two, and absorbing the second, power-saving term, proves (2.4). $\square$

Remark 2.3 (Comparison with the hybrid theorem). Chen–Gupta–Li version 2 [1, Theorem 1.2] states the stronger hybrid estimate

$$\sum_{\chi \bmod q} N(\sigma, T, \chi) \leq (qT)^{o(1)} \left(q^{\frac{7}{3}(1-\sigma)} T^{2(1-\sigma)} + (qT)^{\frac{30}{13}(1-\sigma)} \right).$$

That theorem also implies Proposition 2.2, but it is not used in this paper. Appendix A proves the weaker polylogarithmic-conductor slice needed here from published inputs.

3 Near and far from rational points

Write the center of a short Fourier interval as $\gamma = a/q + \lambda$. If λ is small, Proposition 2.2 controls a neighborhood of a/q . If λ is larger, Matomäki–Radziwiłł–Tao convert the Fourier integral into the two-variable mean from Lemma 2.1. We split at $|\lambda|H_* \asymp R$; both estimates use the same values of a and q .

Proof of Theorem 1.1. Put

$$\rho = \min(\varepsilon/4, 1/1200), \quad H_* = \frac{1}{2}X^{2/15+\rho}, \quad \delta = \rho/16.$$

Every arc of length $1/H$, $H \geq X^{2/15+\varepsilon}$, lies in the concentric arc of length $1/H_*$. It is therefore enough to work at H_* . Set

$$K_0 = \lceil 1/\delta \rceil, \quad \kappa = 2K_0,$$

choose $C_s = C_s(\varepsilon)$ large enough to cover the fixed powers of $\log X$ in the Heath–Brown decomposition and its mean-value estimates, and then take

$$\begin{aligned} B &= \lceil 4(A + C_s + 20) \rceil, \quad Q = L^B, \\ C_c &= A + C_s + 2B + 30, \quad R = L^{C_c}, \quad y = \frac{H_*}{32R}, \\ D_{\text{AP}} &= A + 4B + 10, \quad D = \lceil A + 2B + 10 \rceil, \quad P = L^D. \end{aligned} \quad (3.1)$$

These numbers are fixed before X , the arc length, or the center is chosen. Apply Proposition 2.2 with $\varepsilon_{\text{AP}} = \rho/2$, $K = B$, and $D = D_{\text{AP}}$. After enlarging the lower threshold for X , we have $64L^{C_c} \leq X^{\rho/2}$, and hence $y \geq X^{2/15+\rho/2}$. Thus the proposition applies at length y with exactly the denominator range and logarithmic saving used below.

Near a rational point. For each reduced a/q , introduce the signed measure

$$\nu_{q,a} = \sum_{X < n \leq 2X} \Lambda(n) e(an/q) \delta_n - \frac{\mu(q)}{\varphi(q)} \mathbf{1}_{(X, 2X]}(t) dt.$$

Since $4R/H_* = 1/(8y)$, the signed-measure Gallagher inequality gives

$$\int_{|\beta| \leq 1/(8y)} |\widehat{\nu}_{q,a}(\beta)|^2 d\beta \ll y^{-2} \int_{\mathbb{R}} |\nu_{q,a}((x, x+y])|^2 dx.$$

Decompose the interior sum into reduced residue classes modulo q and apply Cauchy's inequality before Proposition 2.2. The interior, endpoint, and prime-power terms contribute respectively $q^2 X L^{-D_{\text{AP}}}$, $y L^2$, and L^4/y for one rational arc after Gallagher normalization. Summing the reduced numerators over $q \leq Q$ and using $\sum_{q \leq Q} \varphi(q) q^2 \ll Q^4$ and $\sum_{q \leq Q} \varphi(q) \ll Q^2$, while estimating the continuous main term outside $|\beta| \leq P/X$ by $|\beta|^{-1}$, gives on the union $\mathcal{W}$ of intervals $\|\alpha - a/q\| \leq 4R/H_*$, $q \leq Q$,

$$\int_{\mathcal{W} \setminus \mathfrak{M}(P,Q)} |S_X(\alpha)|^2 d\alpha \ll X Q^4 L^{-D_{\text{AP}}} + Q^2 y L^2 + \frac{Q^2 L^4}{y} + \frac{X Q^2}{P} \ll X L^{-A}. \quad (3.2)$$

The last term comes from the continuous main term $\int_X^{2X} e(\beta t) dt$ outside $|\beta| \leq P/X$; prime powers sharing a factor with q and the two endpoints of $(X, 2X]$ give the middle terms. Every term uses the same denominator range $q \leq Q$.

Farther from a rational point. Let γ be the midpoint of a target arc. Dirichlet approximation gives

$$\gamma = a/q + \lambda, \quad q \leq Q, \quad |\lambda| \leq (qQ)^{-1}.$$

If $|\lambda| \leq 2R/H_*$, the target interval lies in $\mathcal{W}$, and (3.2) applies. Otherwise put $U = |\lambda| H_* > 2R$. Corollary 5.3 of Matomäki–Radziwiłł–Tao [8], with $\eta = Q^{-1/2}$, gives

$$\int_{\lambda - 1/H_*}^{\lambda + 1/H_*} |S_X(a/q + \theta)|^2 d\theta \ll \frac{d_2(q)^4}{q U^2} \sup_{q=q_0 q_1} \mathcal{I}(q_0, q_1) + \mathcal{E}_0, \quad (3.3)$$

where

$$\mathcal{I}(q_0, q_1) = \int_I \left(\sum_{\chi \bmod q_1} \int_{t-U}^{t+U} |\mathcal{D}[\Lambda \mathbf{1}_{(X, 2X]}](1/2 + it', \chi, q_0)| dt' \right)^2 dt$$

and I is the range $Q^{-1/2} |\lambda| X \leq |t| \leq Q^{1/2} |\lambda| X$. The denominator in (3.3) is $q U^2$. The remaining term satisfies $\mathcal{E}_0 \ll X L^{O(1)} (Q^{-1/2} + U^{-1})^2$, which is $O(X L^{-A})$ by (3.1).

Use the $m = 8$ Heath–Brown decomposition in [8, Lemmas 2.15–2.16]. Its hypotheses hold because, with $H_0 = H_* X^{-\delta}$,

$$(2/15 + \rho - \delta) - (1/8 + \delta) = 1/120 + 7\rho/8 > 0. \quad (3.4)$$

The small terms save a fixed power of X . The terms with $q_0 > 1$ are supported on powers of the prime dividing q_0 and have squared norm $O(L^3)$; hence $q_0 = 1$ in the remaining convolution terms.

Remove the sharp product cutoff with [8, Corollary 2.5], applied to the *untruncated* convolution f' . Factor the resulting polynomial and then restore the cutoff. The Perron error saves a fixed power of X , and its translations stay inside a fixed enlargement of the same t -range.

Remark 3.1 (Notation in the truncation-removal step). In the corresponding display of [8, Section 6], the occurrences of $\tilde{f}$ in the application of Corollary 2.5 and in the definition of F appear to be typographical slips for the untruncated convolution f' . This interpretation is forced by the preceding identity, Proposition 6.1, and the parallel Appendix A argument. We write the untruncated function explicitly and pay the truncation error above.

The decomposition labels its remaining pieces Type II and Type d_j . Type II and the first two Type- d pieces follow from the character second- and fourth-moment estimates in [8, Lemma 2.10 and Corollary 2.12], as in Section 7 there. After division by qU^2X , the resulting terms are

$$qQ^{1/2}|\lambda|, \quad Q^{1/2}X^{-\delta}, \quad X^{-\delta}, \quad U^{-1}, \quad (3.5)$$

all small in the present range.

For a Type- d_j cell, $3 \leq j \leq 7$, select its shortest displayed β -factor, of length M . At least two other displayed β -factors are no shorter, so the complementary length N' satisfies $N' \gg M^2$. Dyadically decompose the complementary convolution into $O_\varepsilon(1)$ blocks of length N' . On every block, $N' \gg NM_2 \cdots M_j \gg M^2$, and its coefficient remains bounded by a fixed divisor function, at the cost of a fixed logarithmic factor. Cauchy and Fubini bound the overlap of the two inner t' -windows by $O(U)$; expanding the squared character sum leaves at most q^2 mixed means. Lemma 2.1 then gives, after division by qU^2X ,

$$L^{O(1)} \left(qQ^{1/2}|\lambda| + \frac{q}{M} + \frac{q}{U} + \frac{qQ^{1/2}}{H_*} \right) \quad (3.6)$$

which is at most

$$L^{O(1)} \left(Q^{-1/2} + QX^{-2/15-\rho+\delta} + Q/R + Q^{3/2}/H_* \right). \quad (3.7)$$

The first and third terms save powers of L ; the other terms save powers of X . With the values in (3.1), multiplying each term by L^{C_s} still gives $O(L^{-A})$. The definition of C_s includes the number of pieces, both character sums, the Perron shifts, and the divisor factor in (3.3).

The two cases cover every center. In the second case, (3.3) bounds the entire target interval and therefore its minor-arc part. Together with (3.2), this gives (1.1). $\square$

Remark 3.2 (The major arcs must depend on the saving). A rational peak at a denominator r just beyond a fixed cutoff Q has local energy of order X/r^2 , which exceeds XL^{-A} once $A > 2B + O(1)$, and the tail of the continuous main term forces the width exponent to grow with A as well. The order $\forall A \exists B, D$ in Theorem 1.1 is therefore essential; the reversed order $\exists B, D \forall A$ gives a false statement.

4 Prime-pair variance, fourth correlation, and density one

Proof of Theorem 1.3. Let $f(n) = \Lambda(n)\mathbf{1}_{(X, 2X]}(n)$ and $C_f(h) = \sum_n f(n)f(n+h)$. Use the set $\mathfrak{M} = \mathfrak{M}(P, Q)$ supplied by Theorem 1.1, put $\mathfrak{m} = \mathbb{T} \setminus \mathfrak{M}$, and define the major-arc contribution

$$M_f(h) = \int_{\mathfrak{M}} |S_X(\alpha)|^2 e(-h\alpha) d\alpha.$$

A Fejér-kernel large sieve for the positive measure $\mathbf{1}_{\mathfrak{m}}|S_X(\alpha)|^2 d\alpha$ gives

$$\sum_{|h-h_0| \leq H} |C_f(h) - M_f(h)|^2 \ll H \left(\int_{\mathfrak{m}} |S_X|^2 \right) \sup_{|J|=1/H} \int_{J \cap \mathfrak{m}} |S_X|^2. \quad (4.1)$$

Parseval gives $\int_{\mathbb{T}} |S_X|^2 = \sum_{X < n \leq 2X} \Lambda(n)^2 \ll XL$. Use Theorem 1.1 with one extra logarithm to obtain

$$\sum_{|h-h_0| \leq H} |C_f(h) - M_f(h)|^2 \ll HX^2L^{-A}. \quad (4.2)$$

The standard major-arc calculation, using [8, Proposition 4.1], yields, for $0 < |h| < X$,

$$M_f(h) = (X - |h|)\mathfrak{S}(h) + O_C(XL^{-C}\tau(|h|)^2). \quad (4.3)$$

Standard mean-value bounds for fixed powers of τ control the sum of these errors after C is enlarged. The difference between $C_f(h)$ and $R_X(h)$, together with the overlap correction $|h|\mathfrak{S}(h)$, is $O((|h| + 1)L^{O(1)})$ per shift. Since the stated range gives $|h| = O(X^{1-\varepsilon})$, its total square is smaller by a fixed power of X than the right side of (1.2). Adding these contributions to (4.2) gives (1.2).

Write $E_h = R_X(h) - X\mathfrak{S}(h)$. Then

$$\sum_{h \in I} R_X(h)^2 = X^2 \sum_{h \in I} \mathfrak{S}(h)^2 + 2X \sum_{h \in I} \mathfrak{S}(h)E_h + \sum_{h \in I} E_h^2. \quad (4.4)$$

The standard bound $\sum_{h \in I} \mathfrak{S}(h)^2 \ll HL^{O(1)}$, Cauchy's inequality, and (1.2) with a larger logarithmic exponent make the last two terms $O(HX^2L^{-A})$, giving (1.3).

Apply (1.2) with exponent $3A$. Every exception counted in (1.4) contributes more than X^2L^{-2A} to the variance, which gives (1.4). $\square$

5 Averaged Goldbach

Proof of Theorem 1.4. Choose every cutoff independently of the target. For integers $r \geq 1$ define

$$L_X(r) = \mathbf{1}_{X/2 < r \leq X}, \quad Q_X(r) = \mathbf{1}_{1 \leq r < X},$$

$$A_X(N) = \sum_{n+m=N} \Lambda(n)\Lambda(m)L_X(n)Q_X(m), \quad B_X(N) = \sum_{n+m=N} \Lambda(n)\Lambda(m)L_X(n)L_X(m),$$

and

$$E_X(N) = \sum_{n+m=N} \Lambda(n)\Lambda(m)\{1 - L_X(n)Q_X(m) - Q_X(n)L_X(m) + L_X(n)L_X(m)\}.$$

The pointwise indicator identity gives

$$G(N) = 2A_X(N) - B_X(N) + E_X(N). \quad (5.1)$$

Because Q_X excludes X , the overlap in A_X excludes X as well. The endpoint term is

$$2\mathbf{1}_{X/2 < N-X < X} \Lambda(X)\Lambda(N-X)$$

when X is an integer. For sufficiently large X , however, $H \leq X^{1-\varepsilon} \leq X/2$, so this term vanishes. Writing $N = X + h$, the sum $B_X(N)$ and the terms with a summand greater than X each contain $O(h+1)$ pairs. Therefore

$$|B_X(N)| + |E_X(N)| \ll (H+1)L^2 \ll_{A,\varepsilon} XL^{-A}. \quad (5.2)$$

Put

$$S(\alpha) = \sum_{X/2 < n \leq X} \Lambda(n) e(n\alpha), \quad F(\alpha) = \sum_{1 \leq m < X} \Lambda(m) e(m\alpha).$$

Orthogonality gives

$$A_X(N) = \int_{\mathbb{T}} S(\alpha) F(\alpha) e(-N\alpha) d\alpha. \quad (5.3)$$

Let $\mathfrak{M}$ be the union of the rational intervals used in the major-arc calculation below, and put $\mathfrak{m} = \mathbb{T} \setminus \mathfrak{M}$. On $\mathfrak{m}$, define $d\nu = \mathbf{1}_{\mathfrak{m}} S F d\alpha$. For any finite complex measure ν and any block I of at most $H + 1$ consecutive integers, the nonnegative Fejér kernel gives

$$\sum_{k \in I} |\widehat{\nu}(k)|^2 \ll H |\nu|(\mathbb{T}) \sup_{|J|=1/H} |\nu|(J). \quad (5.4)$$

To prove (5.4), shift the frequency block to the origin and use the Fejér kernel $K_{2\lceil H \rceil+2}$, whose triangular Fourier weights are bounded below on that block, including its endpoints. After taking total variation, partition the tail using $K_{2\lceil H \rceil+2}(t) \ll H(1 + H\|t\|_{\mathbb{T}})^{-2}$.

Parseval and Cauchy give $|\nu|(\mathbb{T}) \ll XL$. Apply Theorem 1.1 to S at scale $X/2$ with saving K . Replacing ε by $\varepsilon/2$ makes both bounds

$$(X/2)^{2/15+\varepsilon/2} \leq H \leq (X/2)^{1-\varepsilon/2}$$

hold for large X . Another Cauchy inequality gives

$$\begin{aligned} \sup_{|J|=1/H} |\nu|(J) &\ll XL^{(1-K)/2}. \\ \sum_{X \leq N \leq X+H} |\widehat{\nu}(N)|^2 &\ll HX^2 L^{(3-K)/2}. \end{aligned} \quad (5.5)$$

At the pointwise threshold XL^{-A} , Chebyshev's inequality gives $O(HL^{-A})$ exceptions when

$$K \geq 6A + 3. \quad (5.6)$$

For the major arcs, let $Q = L^B$ and use width L^{B_0}/X . Choose B, B_0 after the exponents defining the MAP arcs at scale $X/2$, and large enough that the present major-arc set contains those arcs. For any fixed $R > 0$, Proposition 4.1 of [8] gives, when $\alpha = a/q + \beta$,

$$\begin{aligned} S(\alpha) &= \frac{\mu(q)}{\varphi(q)} \int_{X/2}^X e(\beta x) dx + O(XL^{-R}), \\ F(\alpha) &= \frac{\mu(q)}{\varphi(q)} \int_0^X e(\beta y) dy + O(XL^{-R}). \end{aligned}$$

For $X \leq N \leq X + H \leq 3X/2$, the continuous convolution has the exact length

$$\int_{\mathbb{R}} \mathbf{1}_{(X/2, X]}(x) \mathbf{1}_{[0, X)}(N - x) dx = X/2. \quad (5.7)$$

After the β integral and the reduced-residue sum,

$$\int_{\mathfrak{M}} S F e(-N\alpha) d\alpha = \frac{X}{2} \mathfrak{S}(N) + O\left(XL^{-C_{\text{major}} \tau(N)^{O(1)}}\right). \quad (5.8)$$

This derivation uses Proposition 4.1 of [8] directly; the displayed Goldbach major-arc proposition there has supports whose continuous overlap vanishes at the displayed coefficient, so it is bypassed. The condition $B \geq 2C_{\text{major}}$ bounds the omitted part of the Ramanujan series, and $B_0 \geq 2B + C_{\text{major}}$ bounds the extension of the β integral. A short-interval mean-value bound for a fixed power of τ , followed by Markov's inequality, gives at most $O(HL^{-A})$ targets on which the error in (5.8) exceeds XL^{-A} .

Choose the parameters in the order

$$A, \varepsilon \rightarrow C_{\text{major}} \rightarrow K \rightarrow B \rightarrow B_0, R \rightarrow X_0.$$

Outside the two exceptional sets, (5.3), (5.5), and (5.8) give

$$A_X(N) = \frac{X}{2} \mathfrak{S}(N) + O_{A,\varepsilon}(XL^{-A}).$$

Equations (5.1)–(5.2) yield $G(N) = X\mathfrak{S}(N) + O(XL^{-A})$. Also, $|(N - X)\mathfrak{S}(N)| \ll HL^{O(1)} \ll XL^{-A}$, after enlarging the input saving, so (1.5) follows.

Every term of $G(N) - W(N)$ contains a proper prime power. Chebyshev's bound gives uniformly

$$|G(N) - W(N)| \ll X^{1/2}L^2,$$

smaller than XL^{-A} for every fixed A . The weighted prime-only formula follows. For even N outside the exceptional set, $\mathfrak{S}(N)$ is bounded below by a positive constant, so $W(N) > 0$ when X is large. $\square$

There is a balanced variant, with both summands in the same dyadic range.

Theorem 5.1 (Balanced averaged Goldbach). *Fix $A, \varepsilon > 0$ and $\delta \in (0, 1/2)$. For an integer $N \in (2X, 4X)$ put*

$$G_X(N) = \sum_{\substack{n+m=N \\ X < n, m \leq 2X}} \Lambda(n)\Lambda(m), \quad \ell_X(N) = \min(N - 2X, 4X - N).$$

Uniformly for $X^{2/15+\varepsilon} \leq H \leq X^{1-\varepsilon}$ and every interval $J \subset [(2+\delta)X, (4-\delta)X]$ of length H ,

$$G_X(N) = \mathfrak{S}(N) \ell_X(N) + O_{A,\varepsilon,\delta}(XL^{-A}) \tag{5.9}$$

for all but $O_{A,\varepsilon,\delta}(HL^{-A})$ integers $N \in J$. Outside the same exceptional set, every even $N \in J$ satisfies

$$\#\{p+q=N : X < p, q \leq 2X\} = \frac{\mathfrak{S}(N) \ell_X(N)}{L^2} \left(1 + O\left(\frac{1}{L}\right)\right).$$

Proof. Both factors are now S_X , so the argument above simplifies. Orthogonality gives $G_X(N) = \int_{\mathbb{T}} S_X(\alpha)^2 e(-N\alpha) d\alpha$. Take $\mathfrak{M}$ as in the major-arc calculation above and put $d\nu = \mathbf{1}_{\mathfrak{M}} S_X^2 d\alpha$. Parseval gives $|\nu|(\mathbb{T}) \leq \sum_{X < n \leq 2X} \Lambda(n)^2 \ll XL$, while Theorem 1.1 with saving K gives directly

$$\sup_{|J'|=1/H} |\nu|(J') \leq \sup_{|J'|=1/H} \int_{J' \cap \mathfrak{M}} |S_X(\alpha)|^2 d\alpha \ll XL^{-K}.$$

No rescaling is needed, because S_X is the sum in Theorem 1.1 itself. Now (5.4) gives

$$\sum_{N \in J} |\widehat{\nu}(N)|^2 \ll HX^2L^{1-K}.$$

Chebyshev's inequality at the threshold XL^{-A} gives $O(HL^{-A})$ exceptions when $K \geq 3A + 1$. On the major arcs, the approximation of S_X by Proposition 4.1 of [8], the β integral, and the reduced-residue sum give, exactly as in (5.8),

$$\int_{\mathfrak{M}} S_X(\alpha)^2 e(-N\alpha) d\alpha = \mathfrak{S}(N) \int_{\mathbb{R}} \mathbf{1}_{(X, 2X]}(x) \mathbf{1}_{(X, 2X]}(N-x) dx + O\left(XL^{-C_{\text{major}}\tau(N)^{O(1)}}\right),$$

and the continuous overlap equals $\ell_X(N)$. The divisor-moment and Markov step is unchanged. Prime powers contribute $O(X^{1/2}L^2)$ uniformly. On J one has $\ell_X(N) \geq \delta X$, and every surviving summand satisfies $\log p \log q = L^2(1 + O(1/L))$, which gives the prime-only count. $\square$

Scope and machine-checked components

The results concern almost every shift or target in every sufficiently long translated interval. A prescribed shift such as $h = 2$, a prescribed Goldbach target, and a pair of prescribed short summand intervals are pointwise problems; the present methods leave them open.

Siegel's theorem enters Proposition 2.2 through the possible exceptional zero, so the constants throughout may be ineffective.

Theorem 1.1 is formalized in Lean 4 as `AllCenterMAP.map_two_fifteenths`. The proof is <https://github.com/jconorgrogan/prime-minor-arcs-2-15> and has been submitted to Palomar, with `comparator.json` targeting that declaration. The release passed a Linux source build of 10,255 jobs, Comparator statement identity against the independently compiled Challenge, independent NanoDa replay, and Lean kernel acceptance. The MAP proof closure contains 100,596 declarations across 1,861 modules; the internal Guth–Maynard large-value closure contains 75,874 declarations. Both closures replayed into an empty kernel. The axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`.

A A published-input density lemma for polylogarithmic conductors

The all-center argument only uses conductors bounded by a fixed power of the logarithm. The following narrower lemma avoids any appeal to an unrefereed hybrid character-aspect theorem while retaining the 30/13 coefficient.

For a Dirichlet character χ , let $N(\sigma, T, \chi)$ count, with multiplicity, the zeros $\rho = \beta + i\gamma$ of $L(s, \chi)$ satisfying $\beta \geq \sigma$ and $|\gamma| \leq T$.

Theorem A.1 (Polylogarithmic-conductor density). *Fix $K > 0$, $\delta > 0$, and $\eta > 0$. Uniformly for*

$$T \geq T_0(K, \delta, \eta), \quad \frac{1}{2} + \delta \leq \sigma \leq \frac{4}{5}, \quad Q \leq (\log T)^K,$$

one has

$$\sum_{q \leq Q} \sum_{\chi \bmod q} N(\sigma, T, \chi) \ll_{K, \delta, \eta} T^{\frac{30}{13}(1-\sigma)+\eta}. \quad (\text{A.1})$$

Proof. We first reduce to primitive characters. If $\chi \pmod{q}$ is induced by the primitive character $\chi^* \pmod{r}$, then

$$L(s, \chi) = L(s, \chi^*) \prod_{\substack{p|q \\ p \nmid r}} (1 - \chi^*(p)p^{-s}).$$

Every zero of a factor in the finite product has real part zero. Consequently the zeros counted in (A.1) are zeros of $L(s, \chi^*)$. The number of character-modulus pairs under consideration is $O(Q^2)$, so it is enough to prove, for every primitive $\chi \pmod{r}$, $r \leq Q$,

$$N(\sigma, T, \chi) \ll_{K, \delta, \eta} T^{\frac{30}{13}(1-\sigma)+\eta/2}. \quad (\text{A.2})$$

We give the reduction of (A.2) to published large-values and moment estimates. This also records where conductor uniformity enters. Fix small constants $\kappa > 0$ and $0 < \vartheta < 1$, to be chosen at the end in terms of η , and put

$$\mathcal{R} = r(T+2), \quad U = 2\mathcal{R}^\kappa, \quad Y = \mathcal{R}^{1/2}.$$

If $r = 1$, the character is the zeta character. The range $7/10 \leq \sigma \leq 4/5$ follows from [3, Theorem 1.2], while the lower range follows from the Ingham estimate summarized in [3, (1.4)]. We may therefore assume $r > 1$; the primitive character χ is then nonprincipal and no pole residue occurs. Form the usual mollifier

$$M_U(s, \chi) = \sum_{d \leq U} \mu(d) \chi(d) d^{-s}.$$

For $\text{Re } s > 1$,

$$L(s, \chi) M_U(s, \chi) = \sum_{n \geq 1} c_n \chi(n) n^{-s}, \quad c_n = \sum_{\substack{d|n \\ d \leq U}} \mu(d). \quad (\text{A.3})$$

Thus $c_1 = 1$ and $c_n = 0$ for $1 < n \leq U$. We now record the detector without discarding its weights. Mellin inversion for $e^{-n/Y}$ and a contour shift to $\text{Re } z = 1/2 - \beta$ give, at a zero $\rho = \beta + i\gamma$,

$$e^{-1/Y} + \sum_{n > U} c_n \chi(n) n^{-\rho} e^{-n/Y} = \frac{1}{2\pi i} \int_{\text{Re } z = 1/2 - \beta} \Gamma(z) Y^z L(\rho + z, \chi) M_U(\rho + z, \chi) dz. \quad (\text{A.4})$$

The apparent pole at $z = 0$ is removable: since $L(\rho, \chi) = 0$, $L(\rho + z, \chi) = z H_\rho(z)$ with H_ρ holomorphic near zero, while $z\Gamma(z) = \Gamma(z+1)$. There is no pole at $z = 1 - \rho$ because χ is nonprincipal. If $N_\chi(t, t+1)$ denotes the number of nontrivial zeros in the critical strip, counted with multiplicity, whose ordinates lie in $[t, t+1]$, we use the local zero count

$$N_\chi(t, t+1) \ll \log(r(|t|+2)). \quad (\text{A.5})$$

This estimate controls multiplicity. Put $a = 1/2 - \beta$ and $B = (\log \mathcal{R})^2$. Parametrizing the shifted line by $z = a + iv$ gives $dz = i dv$ and $\rho + z = 1/2 + i(\gamma + v)$. Equivalently, with the absolute critical-line ordinate $u = \gamma + v$, the integrand is

$$\Gamma(a + i(u - \gamma)) Y^{a+i(u-\gamma)} L(1/2 + iu, \chi) M_U(1/2 + iu, \chi),$$

and the truncated interval is $|u - \gamma| \leq B$. Uniformly for $1/2 < \beta \leq 1$ and $|v| \geq 1$, the fixed-strip bound for L , the elementary bound $|M_U(1/2 + i(\gamma + v), \chi)| \leq U + 1$, and vertical Gamma decay give

$$|\Gamma(a + iv) Y^{a+iv} L(1/2 + i(\gamma + v), \chi) M_U(1/2 + i(\gamma + v), \chi)|$$

$$\begin{aligned}
&\ll r^2(U+1)(1+|v|)(5+|\gamma|+|v|)^2e^{-|v|} \\
&\ll r^2(U+1)(5+|\gamma|)^2(1+|v|)^3e^{-|v|}.
\end{aligned}$$

Hence the two vertical tails are

$$\ll r^2(U+1)(5+|\gamma|)^2e^{-B/2}.$$

If $N = \lceil YB \rceil$, the arithmetic tail is at most

$$(U+1)e^{-(N+1)/Y}(1-e^{-1/Y})^{-1} \ll (U+1)Ye^{-B}.$$

Since $\mathcal{R} = r(T+2)$, $Y = \mathcal{R}^{1/2}$, $U = 2\mathcal{R}^\kappa$, $|\gamma| \leq T$, and $r \leq (\log T)^K$, both tails are $O_A(\mathcal{R}^{-A})$ for every fixed A . Consequently every remaining zero satisfies either

$$\left| \sum_{U < n \leq Y(\log \mathcal{R})^2} c_n \chi(n) n^{-\rho} e^{-n/Y} \right| \gg 1, \quad (\text{I})$$

or the shifted critical-line integral containing $L(1/2 + i(\gamma + u), \chi) M_U(1/2 + i(\gamma + u), \chi)$ is large for $|u| \leq (\log \mathcal{R})^2$. This is the classical Montgomery Dirichlet- L detector [4, Chapters 10 and 12]; (A.3) displays the character-twisted coefficient identity. Equation (A.5) controls multiplicity and permits thinning at any prescribed separation, with the separation length times a logarithm as the loss.

For the critical-line alternative, first thin the original ordinates at separation $3(\log \mathcal{R})^2$. The shifted ordinates are then one-separated, and the thinning loss is polylogarithmic by (A.5). The detector inequality and the integrability of the gamma factor imply, for one shifted ordinate v , that

$$|L(1/2 + iv, \chi) M_U(1/2 + iv, \chi)| \gg Y^{\sigma-1/2} \mathcal{R}^{-o(1)}.$$

Since

$$|M_U(1/2 + iv, \chi)| \leq \sum_{d \leq U} d^{-1/2} \ll U^{1/2},$$

Montgomery's discrete family fourth moment [4, Theorem 10.3], restricted to the fixed primitive character by nonnegativity,

$$\sum_v |L(1/2 + iv, \chi)|^4 \ll (rT)^{1+o(1)}$$

at one-separated ordinates gives

$$\#\mathcal{Z}_{II} \ll \mathcal{R}^{2(1-\sigma)+2\kappa+o(1)}. \quad (\text{A.6})$$

All conductor factors are harmless because $r \leq (\log T)^K$. Notice that we used the pointwise bound for M_U ; a fourth moment of L and only a second moment of M_U would not justify (A.6).

It remains to count the zeros satisfying (I). Dyadic subdivision supplies one N with

$$\mathcal{R}^\kappa \ll N \ll \mathcal{R}^{1/2}(\log \mathcal{R})^2 \quad (\text{A.7})$$

and a large block for a proportion $(\log \mathcal{R})^{-O(1)}$ of these zeros. The lower bound holds because the coefficients in (A.3) vanish for $1 < n \leq U$. We retain the exponential weight in the coefficients. The real part of a zero must also be removed before applying one large-values theorem to the

whole set. Fix the left edge σ , put $L = \log N$, choose a fixed $\zeta_0 \in C_c^\infty((1/2, 2))$ equal to one near $[1, 1 + \log(2)/L]$, and set

$$\psi_\beta(u) = \zeta_0(u/L)e^{-(\beta-\sigma)u}.$$

Uniformly for $\sigma \leq \beta \leq 1$,

$$\|\psi_\beta^{(j)}\|_1 \ll_j L^{1-j}, \quad |\widehat{\psi}_\beta(\xi)| \ll_j L^{1-j}|\xi|^{-j}, \quad \|\widehat{\psi}_\beta\|_1 \ll 1. \quad (\text{A.8})$$

Fourier inversion gives, up to the fixed Fourier-sign convention,

$$\sum_{N < n \leq 2N} c_n e^{-n/Y} \chi(n) n^{-\beta-i\gamma} = \int_{\mathbb{R}} \widehat{\psi}_\beta(\xi) \sum_{N < n \leq 2N} c_n e^{-n/Y} \chi(n) n^{-\sigma-i(\gamma+2\pi\xi)} d\xi. \quad (\text{A.9})$$

Put $H = \mathcal{R}^\vartheta$. By (A.8), choosing the integration-by-parts order in terms of ϑ makes the contribution of $|\xi| > H$ smaller than the block threshold, uniformly in β . Indeed, $|c_n| \leq d(n)$ gives the crude uniform estimate

$$\sup_t \left| \sum_{N < n \leq 2N} c_n e^{-n/Y} \chi(n) n^{-\sigma-it} \right| \ll \mathcal{R}^{o(1)} N^{1-\sigma},$$

and its product with the Fourier tail from (A.8) is smaller than any prescribed fixed power of $\mathcal{R}^{-1}$ after increasing the integration-by-parts order. Thus every retained zero chooses some $|\xi_\rho| \leq H$ where a common polynomial is large. Normalize with

$$C_N = \max_{N < n \leq 2N} |c_n| = \mathcal{R}^{o(1)}$$

(the block is nonzero), and define the zero-independent polynomial

$$D_\chi(t) = \sum_{N < n \leq 2N} a_n \chi(n) n^{it}, \quad a_n = \frac{c_n e^{-n/Y} (N/n)^\sigma}{C_N}, \quad |a_n| \leq 1, \quad (\text{A.10})$$

which satisfies

$$|D_\chi(t_j)| \geq N^\sigma \mathcal{R}^{-o(1)} \quad (\text{A.11})$$

at a shifted ordinate $t_j = -\gamma - 2\pi\xi_\rho$. Any unit interval contains at most $O((H+1)\log \mathcal{R})$ of these shifted ordinates, counted with multiplicity: all their original ordinates lie in an interval of length $O(H)$, so this follows from (A.5). Greedy selection therefore leaves a one-separated subset of cardinality

$$\gg \#\mathcal{Z}_I / ((H+1)\log \mathcal{R}). \quad (\text{A.12})$$

It lies in an interval of length $O(T+H) = O(T)$. Translating that interval multiplies the common coefficients by unimodular factors. This quantifies the real-part-removal step compressed in [3, Section 13.1]; the fixed factor $\chi(n)e^{-n/Y}$ remains in the coefficients throughout.

Guth and Maynard's published large-values theorem [3, Theorem 1.1] applies to every complex coefficient sequence bounded by one. Hence it applies to (A.10). Their power argument in [3, Section 13.1] has two branches: Theorem 1.1 is used when the powered length is at most their switching scale, and the usual discrete mean-value theorem is used in the complementary branch. Both inputs allow arbitrary bounded complex coefficients. In the present twisted setting, complete multiplicativity gives

$$\chi(n_1) \cdots \chi(n_k) = \chi(n_1 \cdots n_k),$$

and, literally,

$$D_\chi^k(t) = \sum_{N^k < m \leq (2N)^k} \chi(m) b_m m^{it}, \quad |b_m| \leq d_k(m) = \mathcal{R}^{o(1)}.$$

Split this support into $O_k(1)$ dyadic blocks. The triangle inequality and pigeonholing retain one common block for a fixed positive proportion of the selected ordinates; normalize its divisor-bounded coefficients before applying [3, Theorem 1.1]. These operations cost only $\mathcal{R}^{o(1)}$. By (A.7) and $r \leq (\log T)^K$, the polynomial length obeys $N \leq T^{1/2+o_K(1)}$ and $N \geq T^{\kappa+o_K(1)}$. The integer k in their proof is therefore bounded in terms of κ ; this is enough because κ is fixed before $T \rightarrow \infty$. The same power argument gives, for $7/10 \leq \sigma \leq 4/5$,

$$\#\mathcal{Z}_I \leq T^{\frac{15(1-\sigma)}{3+5\sigma} + \vartheta + o_{K,\kappa}(1)}. \quad (\text{A.13})$$

Because

$$\frac{15}{3+5\sigma} \leq \frac{30}{13} \quad \left(\sigma \geq \frac{7}{10} \right), \quad \min_{7/10 \leq \sigma \leq 4/5} \left(\frac{15}{3+5\sigma} - 2 \right) (1-\sigma) = \frac{1}{35},$$

we may first choose κ so that $2\kappa < 1/70$ and then choose ϑ and all remaining subpower losses smaller than the allocated $\eta/2$. Equations (A.6) and (A.13) prove (A.2) in this range.

For $1/2 + \delta \leq \sigma \leq 7/10$, Montgomery's classical family estimate [4, Theorem 12.1] gives, for each modulus r ,

$$\sum_{\chi \bmod r} N(\sigma, T, \chi) \ll (rT)^{\frac{3(1-\sigma)}{2-\sigma}} (\log(rT))^9; \quad (\text{A.14})$$

for the fixed character under consideration, nonnegativity gives explicitly

$$N(\sigma, T, \chi) \leq \sum_{\psi \bmod r} N(\sigma, T, \psi).$$

Since

$$\frac{3}{2-\sigma} \leq \frac{30}{13} \quad \left(\sigma \leq \frac{7}{10} \right),$$

(A.2) follows throughout the asserted strip after taking T sufficiently large in terms of K, δ, η .

Finally $Q^{O(1)} = (\log T)^{O_K(1)} = T^{o_K(1)}$. Summing (A.2) over the $O(Q^2)$ original character-modulus pairs and allocating the total subpower loss to $\eta/2$ proves (A.1). $\square$

Remark A.2 (Scope of the density lemma). The theorem does not extend the Guth–Maynard estimate to a character family. It applies that published arbitrary-coefficient theorem separately to each character and uses that the family is only polylogarithmic. The fixed-character detector, fourth-moment input, real-part removal, spacing loss, and bounded-power coefficient normalization are displayed above because they are the uniformity points on which this reduction depends.

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